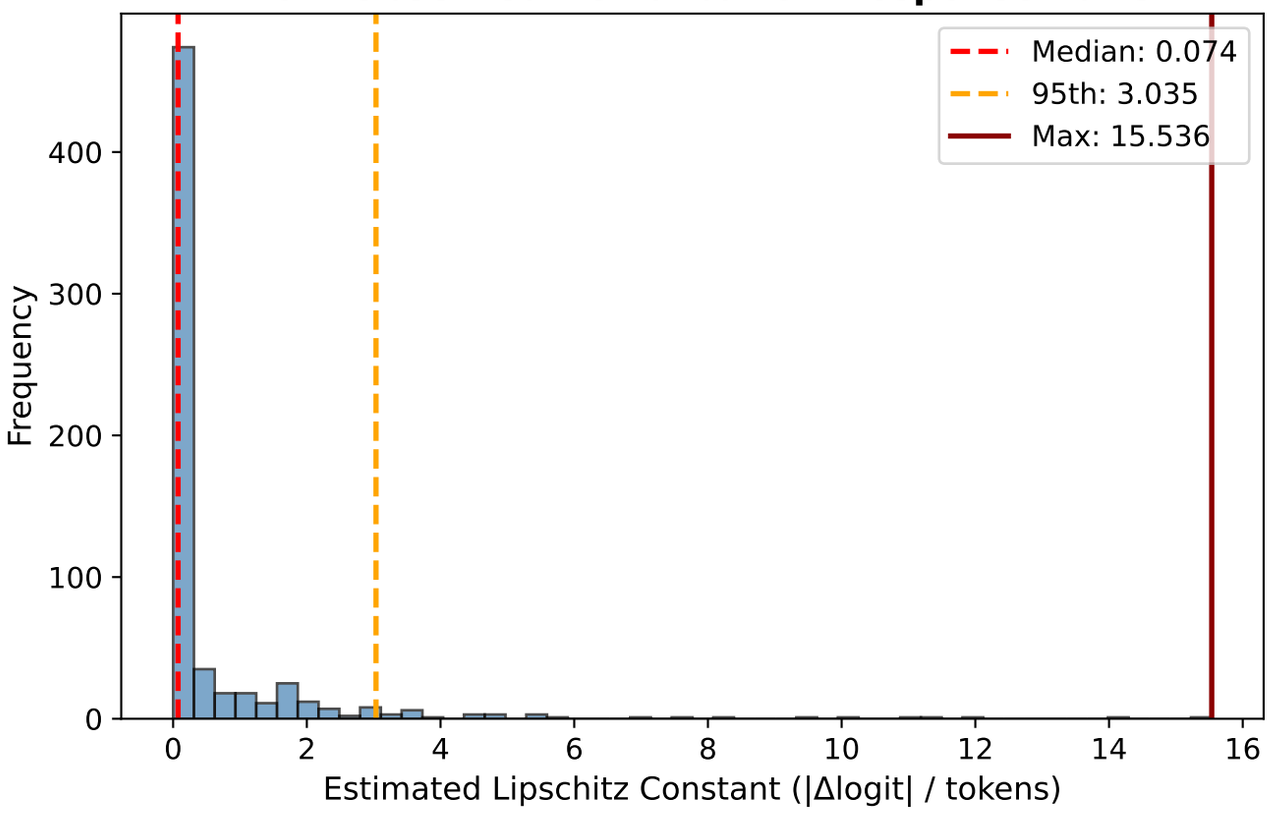
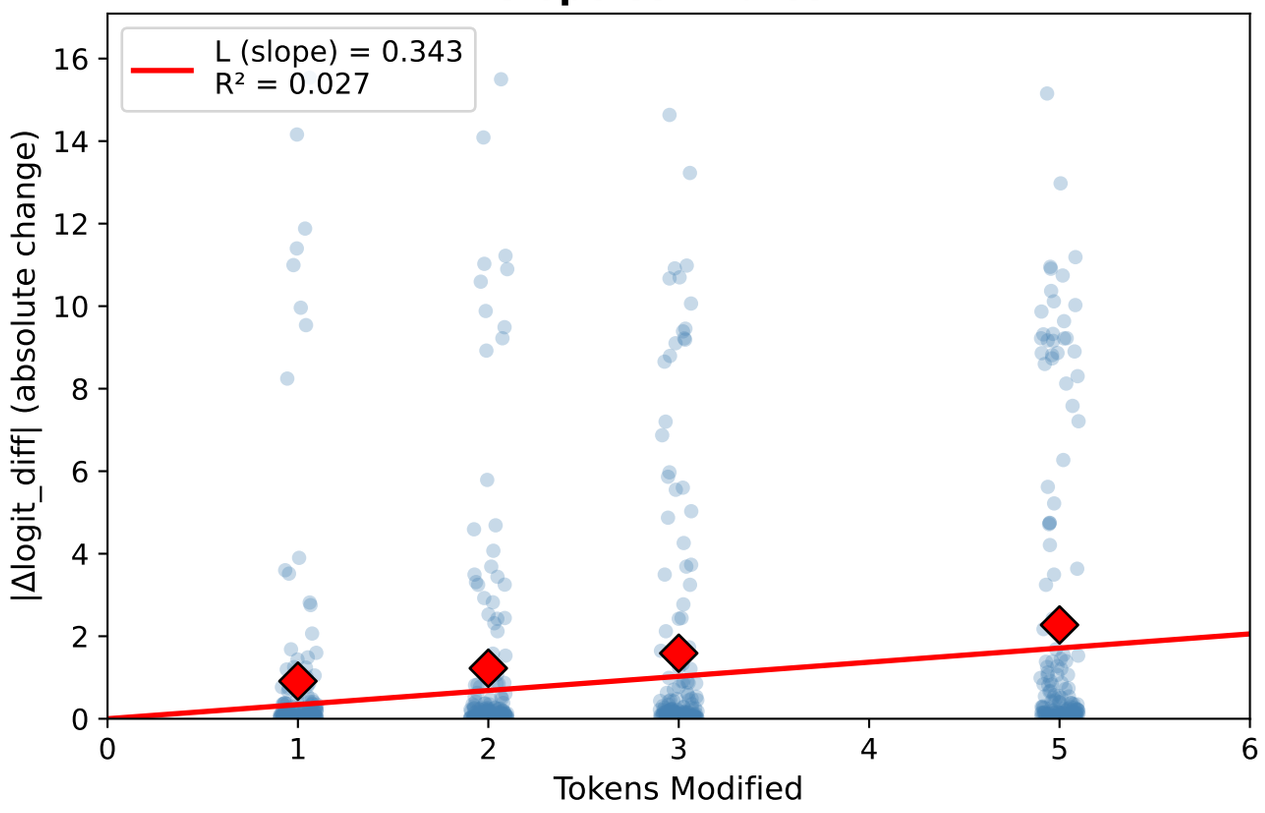
Lipschitz Constant Estimation: Completing the Adversarial Margin Formula

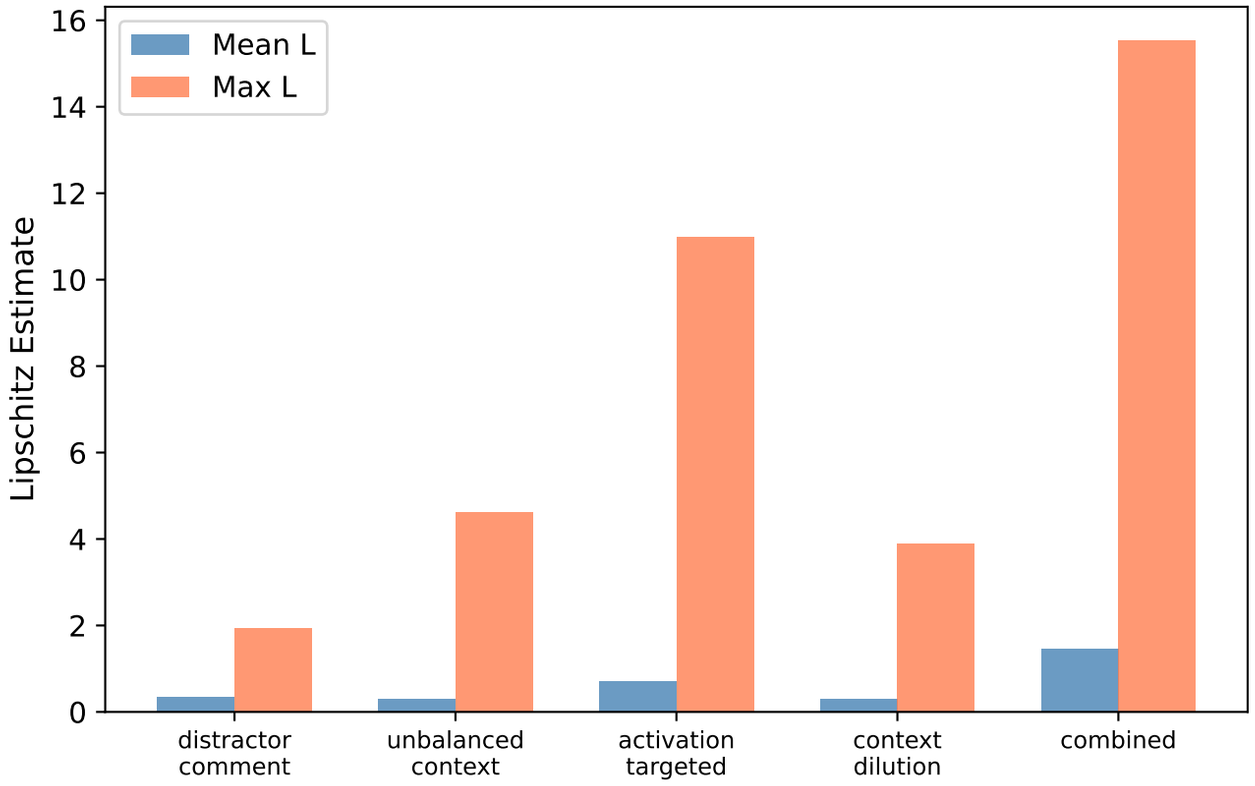
A. Distribution of Lipschitz Estimates From Observed Perturbation-Response Pairs



B. Lipschitz Scaling: Logit Change vs Perturbation Slope estimates L



C. Lipschitz Estimates by Attack Strategy Different strategies have different effective L




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                LIPSCHITZ ESTIMATION RESULTS
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ESTIMATED VALUES
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L (max observed):     15.5361
L (99th percentile):  9.0345
L (95th percentile):  3.0351  ← RECOMMENDED
L (median):           0.0737
L (regression slope): 0.3428 (R² = 0.027)

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                COMPLETE MARGIN FORMULA
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Using L = 3.0351 (95th percentile, conservative):

                    |g(v)|              |logit_diff|
margin = ──────────────────────  = ──────────────────────
          L × (1 + K)              3.0351 × (1 + K)

For token involution (K ≈ 1):

margin ≈ |logit_diff| / 6.0702

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                INTERPRETATION
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• The Lipschitz constant bounds how much the model's output
  can change per unit of input perturbation

• L ≈ 3.04 means each token modification can shift
  the logit_diff by at most ~3.04 units

• This confirms the model has bounded sensitivity to inputs,
  validating a key assumption of the Collision Principle

• Involution-targeted attacks (activation_targeted) achieve
  highest effective L, confirming they exploit circuit structure

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